



Gantla Venkata Sravan

Roll No.: EE24B013

B.Tech. (Electrical Engineering)

Indian Institute of Technology, Tirupati

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (EE)	Indian Institute of Technology, Tirupati	8.29	2024–2028
Senior Secondary (XI–XII)	Board of Intermediate Education, Andhra Pradesh	92.5%	2022–2024
Secondary (X)	Board of Secondary Education, Andhra Pradesh	92.5%	2022

POSITIONS OF RESPONSIBILITY

- **Head, Avionics Division**, Rocketry Club (VelR), IIT Tirupati *Aug. 2025 – Present*
- **Core Member, CubeSat Team**, Gagan Vedhi, IIT Tirupati *Aug. 2025 – Present*
- **Core Member, Rocketry Project**, Gagan Vedhi, IIT Tirupati *Feb. 2025 – May 2025*

PROJECTS

- **Wi-Fi Enabled Custom DAQ & Control Unit (v1 & v2)** *Sept. 2025 – Feb. 2026*
Avionics Division, Rocketry Club (VelR), IIT Tirupati [GitHub](#)
 - **Tools:** C/C++, Arduino Mega, PlatformIO, Embedded Systems, PCB Design, Bare Metal Programming
 - Designed and fabricated two iterations of a Wi-Fi-enabled DAQ and control unit for solid rocket motor static fires, with rigorously tested hardware and software fail-safes for safe ignition and abort sequences.
- **Avionics Board v1** *Feb. 2025 – May 2025*
Rocketry Club (VelR), IIT Tirupati [GitHub](#)
 - **Tools:** C/C++, Arduino Mega, PlatformIO, PCB Design, Embedded Systems
 - Sole schematic author for the club's first avionics board, designed before assuming the Avionics Head role.
- **Modular High-Performance Avionics Board v2** *Jan. 2026 – Mar. 2026*
Avionics Division, Rocketry Club (VelR), IIT Tirupati [GitHub](#)
 - **Tools:** C/C++, Teensy 4.1, PlatformIO, PCB Design, Embedded Systems, Bare Metal Programming
 - Designed and fabricated a modular avionics board with fail-safes, flight data logging, apogee detection, and parachute deployment; hardware architecture designed to support active flight control surfaces and thrust vector control.

PERSONAL PROJECTS

- **CHOTU – Self-Balancing Two-Wheeled Robot** *2024 – Present*
Personal Project [GitHub](#)
 - **Tools:** MicroPython, Embedded Systems, Control Systems, Wi-Fi
 - Self-balancing robot with cascaded PID control; ongoing with incremental hardware upgrades, now including Wi-Fi telemetry and remote control.
- **STM32 Bare Metal Custom Library** *Jul. 2025 – Present*
Personal Project / Learning
 - **Tools:** C, STM32, Bare Metal Programming
 - Custom bare metal library for STM32 covering clock tree configuration, GPIO, and timers – no vendor HAL.
- **STM32 USB-HID Emulator** *Mar. 2026*
Personal Project
 - Implemented a USB HID device on STM32 that enumerates as a keyboard or mouse on a host PC; involved deep study of the USB specification to understand enumeration, descriptors, and HID protocol.
- **nRF52 Health Monitoring System** *Mar. 2026 – Present*
Freelance Project
 - **Tools:** C, Zephyr RTOS, nRF52, Bluetooth LE
 - Health monitoring system on nRF52 with Zephyr RTOS; sensor data transmitted to a mobile client over BLE notifications.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, MicroPython, Python, C#
- **Embedded & Hardware:** Bare Metal Programming & Debugging, Zephyr RTOS, PCB Design, STM32, nRF52
- **Tools & OS:** Git, PlatformIO, Ubuntu/Linux, BusyBox, MATLAB, LTspice

OTHER EXPERIENCE

- **Game Developer** – High School, 2020–2022
 - Developed games in Unity (C#); freelance work, team collaboration, and industry exposure. No titles commercially published. [\[Gameplay Video\]](#)